INCLUSIVE FORUM ON
CARBON MITIGATION APPROACHES
PAPERS

The IFCMA's Climate Policy Database

Policy Instruments Typology and Data Structure

BETTER POLICIES FOR BETTER LIVES

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Policy Instruments Typology and Data Structure

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Background

The Inclusive Forum on Carbon Mitigation Approaches is the OECD's flagship initiative to help optimise the global impact of emissions reduction efforts around the world through better data and information sharing, evidence-based mutual learning, and inclusive multilateral dialogue.

By taking stock of different carbon mitigation approaches, mapping policies to the emissions they cover, and estimating their comparative impact in terms of emissions reductions, the IFCMA is enhancing understanding of the effect of the full spectrum of carbon mitigation approaches deployed around the world and their combined global impact. The IFCMA is also identifying and addressing challenges related to the calculation of sector- and product-level carbon intensity metrics, relevant to the design and evaluation of mitigation policies, and to steer firms' and consumers' decisions towards lower-emission products. This work supports better international coordination to avoid the proliferation of different standards, help minimise compliance costs for business, and avoid disruptions to trade.

To advance its technical work, the IFCMA brings together delegates from the climate, tax, and structural economic policy communities from more than 55 IFCMA members and numerous countries participating as Invitees around the world.

Inclusive Forum on Carbon Mitigation Approaches Papers

The IFCMA Papers series brings together outputs from the initiative's work to take stock of different carbon mitigation approaches, map policies to the emissions they cover, and estimate their impact on greenhouse gas emissions, as well as its work on analysing methodologies for computing the carbon intensity of goods and sectors. Comments on IFCMA Papers are welcome at IFCMA@oecd.org.

Executive Summary

This paper presents a framework for a comprehensive database of climate change mitigation policy instruments, developed by the Inclusive Forum on Carbon Mitigation Approaches (IFCMA). The database aims to systematically classify and describe policy instruments across member countries, enabling detailed analysis and comparison. The data structure, building on extensive literature and pilot studies, includes a clear typology categorising instruments by their operating mechanisms (e.g., economic, regulatory, information instruments). It also incorporates standardised attributes to characterise policies, supporting a harmonised stocktake. The framework aligns with international reporting standards on climate policy action. It aims to enhance the understanding of diverse mitigation approaches, supporting global climate goals, while opening new avenues for empirical research on climate action.

Keywords: environment, environmental statistics, accounts and indicators, climate change mitigation, policy instruments, policy reporting, policy database.

JEL classification codes: H: Public economics; K32 Environmental, Energy, Health, and Safety Law; Q01 Sustainable Development; Q480 Energy: Government Policy; Q54 Climate; Natural Disasters and Their Management; Global Warming; Q580 Environmental Economics: Government Policy.

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1 Introduction

Since the adoption of the Paris Agreement, countries have significantly expanded their climate actions, setting more ambitious targets and implementing a range of policy instruments tailored to their national circumstances and objectives (IPCC, 2023[1]; OECD, 2023[2]). To provide a comprehensive overview of progress towards national and global targets, it is essential to identify and understand countries' policy choices.

The Inclusive Forum on Carbon Mitigation Approaches (IFCMA) is undertaking a systematic stocktake of countries' climate change mitigation and mitigation-relevant policy instruments.[1] The corresponding database, which will encompass policy information for all IFCMA members, represents a primary technical output of the IFCMA. The IFCMA's work plan envisions the first version of this database to be delivered by the end of 2025, with regular updates thereafter.

Constructing a harmonised policy database involves two key components: i) a data structure for the coherent organisation of data, and ii) an effective and efficient data collection, management, and validation process. This document focuses on the former. First, it proposes a clear typology, based on an extensive literature review, to group policy instruments into consistent categories (Section 2). Second, it outlines a comprehensive framework (i.e. the data structure) to systematically describe climate policy instruments across their multiple dimensions. This includes, for example, precise definitions of the units of analysis and standardised variables (referred to as policy "attributes") to characterise instruments (Section 3). A separate document will present the approach and infrastructure requirements for an efficient and effective data collection, management, and validation process.

The data structure is more comprehensive than those of existing databases, covering a broader policy scope and including more and more precisely defined policy attributes. It aims for allowing a harmonised stocktake of mitigation policies, complementing bottom-up processes such as the Enhanced Transparency Framework (ETF) under the Paris Agreement. This, in turn, will support the understanding and comparative analysis of different mitigation approaches, and enable new empirical analysis. Additionally, the database will include information on an instrument's greenhouse gas (GHG) emission base – a new indicator offering insights into the emissions that a policy instrument covers through its policy structure (OECD, 2024[3]).

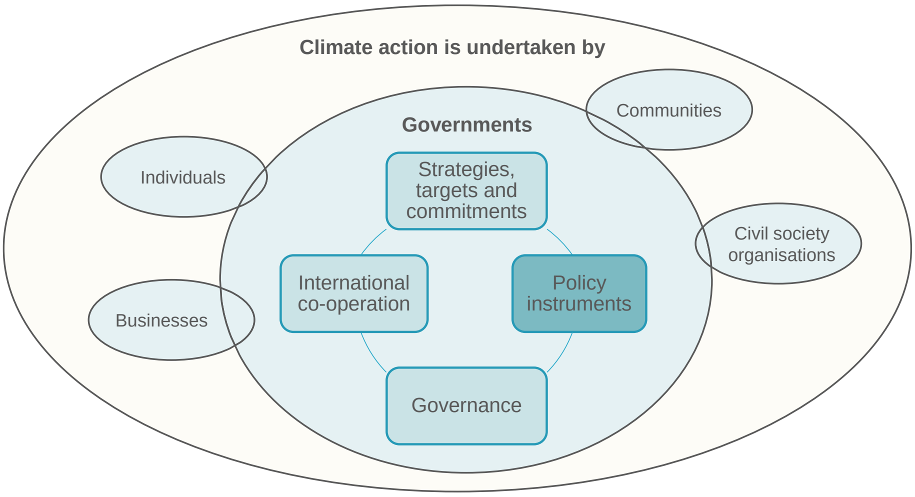
The typology and data structure build on early findings from IFCMA pilot studies on Switzerland and Chile, as well as experience acquired from existing OECD policy databases such as the OECD Policy Instruments for the Environment (PINE) Database, the Climate Actions and Policies Measurement Framework (CAPMF), the OECD Inventory of Support Measures for Fossil Fuels (FFS), and data collected for the OECD Carbon Pricing and Energy Taxation (CPET) dataset. Both the typology and data structure are fully coherent with data reporting structures of the ETF of the United Nations Framework Convention on Climate Change (UNFCCC) and definitions and concepts used by the Intergovernmental Panel on Climate Change (IPCC). It is important to note, however, that the structure goes beyond what the first version of the IFCMA policy database of 2025 will entail, and the framework in this note may evolve further with experience from additional IFCMA pilot studies.

[1] This document refers to "climate change mitigation" policy instruments as those instruments that explicitly intend to reduce or remove GHG emissions. "Mitigation-relevant instruments" refer to instruments that may induce changes in emissions (positive or negative) without necessarily having emission reductions or removals as a primary goal. Hereafter, "climate policy instruments" will refer to both climate change mitigation and mitigation-relevant instruments.

2 Typology of climate change mitigation policy instruments

Developing a typology for climate policy instruments begins with defining the unit of observation, a challenging task given the multitude of forms climate actions can take. Climate action can be driven by various stakeholders (governments at all levels, businesses, communities, individuals, etc.) and can manifest in different ways. For instance, climate action led by governments includes establishing strategies, targets and commitments, policy instruments for their implementation, as well as effective governance structures and international co-operation (Figure 2.1).

Figure 2.1. Dimensions of climate action



Source: Authors.

The IFCMA's work plan established that the policy stocktake should focus on climate change mitigation and climate change mitigation-relevant **policy instruments**. Policy instruments are defined as tools through which governments encourage, mandate, or guide behaviour to achieve specific objectives by creating rules, offering services, reallocating resources, or providing information. Most policy instruments impact economic agents – individuals or entities – by establishing enforceable legal obligations (e.g. complying with a standard or paying a tax) or providing legal avenues for engaging in or benefitting from specific activities (e.g. receiving a subsidy). Policy instruments differ from policy strategies, targets and commitments, which guide behaviour but do not impose rules, benefits and costs on specific agents.

Policy instruments can take various forms. To facilitate understanding and analysis, multiple **typologies** have been proposed to group or classify these instruments into specific, pre-defined categories or classes based on common characteristics, attributes, or criteria. A policy typology is crucial not only for improving database accessibility – allowing users to filter and focus on specific subsets of information – but also for supporting consistent, robust policy analysis. By classifying policies systematically, typologies help in structuring comparative assessments and identifying patterns across different context. Moreover, a well-defined policy typology reduces the risk of inconsistency or ambiguity in policy discussions, ensuring that policies are interpreted consistently. Numerous policy typologies have been proposed in the literature to classify environmental or climate policy instruments. While many of them adopt a similar approach, they vary in the level of detail they cover and specific definitions, reflecting their distinct purposes.

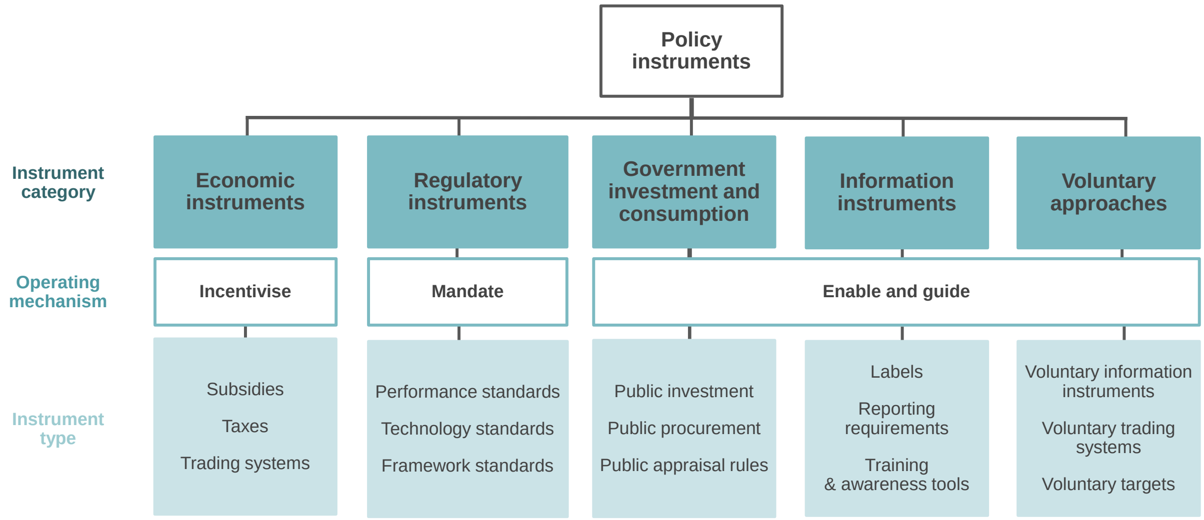
This section outlines the **policy instrument typology for the IFCMA Climate Policy Database**. Establishing a clear, comprehensive and consistent typology for climate change mitigation policy instruments within the IFCMA is critical to support policy dialogue and minimise the risk of inconsistency or ambiguity in policy discussions and analyses. The typology builds on an extensive review of existing typologies, including those introduced by international organisations such as OECD, IPCC, and the European Commission. However, the IFCMA typology extends these typologies to address shortcomings. Specifically, the IFCMA typology aims to create a comprehensive, consistent, and clearly defined typology capable of systematically, coherently, and transparently classifying the breadth of mitigation approaches adopted by IFCMA members. Annex A discusses the most pertinent typologies analysed, their respective relevance for the IFCMA and their shortcomings for the IFCMA's purposes.

The IFCMA's typology is a functional classification based on the instrument's **operating mechanism**. This refers to the rules and processes that govern the instrument's implementation, closely linked to the legal obligation imposed on the regulated economic agents and the expected behavioural change resulting from them. For example, some instruments may operate by providing financial incentives to reward or encourage desired behaviours, while others rely on mandates or restrictions, and some operate by providing information. This classification framework, known as "carrots, sticks and sermons" (Bemelmans-Videc, Rist and Vedung, 1998[4]), has formed the foundation for numerous typologies in the literature, including those used by the IPCC (Dubash et al., 2022[5]) and the UNFCCC reporting framework.

Based on the operating mechanism, the IFCMA typology distinguishes five major categories: **economic** instruments, **regulatory** instruments, **government** investment and consumption, **information** instruments, and **voluntary** approaches (Figure 2.2). Each of those categories is further separated into distinct policy types. Additionally, each observation will be classified as one specific instrument – such as "fuel economy standard" – regardless of varying names like fuel economy, fuel efficiency, minimum energy efficiency, or CO_2 standard. This standardised classification allows for easier filtering, searching, and analysis of policies across the database. Examples for instruments are provided in Table 1. The following section outlines the IFCMA's main categories and their respective instrument types. Precise definitions of each category and group, along with clarifications and examples, are provided in Annex B.

Since the typology is based on the operating mechanism, it may overlook classifications that are relevant for specific analytical objectives. For instance, it may not clearly capture nudging policies as these could operate through different mechanisms (e.g. regulatory, information or economic instruments). However, the typology, along with the policy attributes outlined in Section 3, is designed to be flexible, allowing for the development of additional policy categories and groups or reorganisation of data to meet specific analytical needs. Additionally, it is important to note that the typology currently focusses exclusively on policy instruments and therefore does not capture other forms of climate action, such as government strategies, targets or commitments, international co-operation mechanisms, or private sector-led initiatives. Future work could potentially expand the typology to encompass these broader aspects of climate action.

Figure 2.2. Policy typology for IFCMA



Source: Authors.

Economic instruments

Economic instruments refer to instruments that incentivise specific behaviour through a direct monetary incentive on the levels of production or consumption, influencing the prices or levels of production or consumption at the margin. The IFCMA proposes to further distinguish economic instruments into three instrument types: **subsidies**, **taxes** and **trading systems** (see Annex B).

Regulatory instruments

Regulatory instruments refer to instruments that restrict or mandate specific actions or outcomes with respect to economic agents, for example by setting limits, or define allowable actions to promote the adoption of desired processes, technologies, products, or outcomes. These instruments can take the form of direct mandates, prohibitions, or restrictions on specific goods or activities, establish maximum or minimum limits on various parameters (such as emissions or emission intensities), or require agents to obtain explicit permission or meet specific criteria before engaging in certain activities. The IFCMA proposes to further categorise regulatory instruments into three instrument types: **performance standards**, **technology standards**, and **framework standards** (see Annex B).

Government investment and consumption

Government investment (e.g. in infrastructure or research and development) and consumption instruments guide, limit or require action by the government itself, setting an example or creating favourable conditions for or guiding economic agents. The IFCMA proposes to further categorise government instruments into three instrument types: **public investment**, **public procurement** and **public appraisal rules** (i.e. project and programme evaluation rules imposed on the government) (see Annex B).

Information instruments

Information instruments mandate the collection and/or dissemination of information to raise awareness and communicate knowledge to influence behaviour. The IFCMA proposes to further categorise information instruments into **reporting requirements**, where regulated agents are mandated to provide environmental information; and **capacity building and awareness** where the government provides information to agents (see Annex B).

Voluntary approaches

Voluntary approaches refer to agreements between a government authority and one or more private parties to achieve climate objectives or to improve performance beyond what is required by regulations. They may also include frameworks established by governments to facilitate private action or to test instruments before they become mandatory (e.g. voluntary energy efficiency labels that firms may decide to use). The IFCMA proposes to further categorise voluntary approaches into **voluntary information instruments**, **voluntary GHG emission trading systems**,[2] and **voluntary targets** such as GHG emission reductions or energy efficiency improvements) (see Annex B).

Table 1. Examples of policy instruments, classified along IFCMA typology

Instrument category	Instrument type	Instrument examples
Economic instruments	Subsidies	Renewable electricity feed-in tariff Subsidy for carbon capture, use and storage technologies Vehicle purchase subsidy Payment for eco-system services
	Taxes & tax incentives	Carbon tax Fuel excise tax Corporate income tax allowance for clean technologies
	Trading systems	GHG emission trading system Tradable renewable energy quotas
Regulatory instruments	Performance standards	GHG emissions standard Fuel economy standard Minimum energy performance standard for appliances, buildings, etc.
	Technology standards	Biofuel quota Electric vehicle sales mandate Bans/ phase out of coal-fired power generation
	Framework standards	Requirement to adopt an energy management system (e.g. ISO 50001) Electricity marked regulations (e.g. priority grid access/dispatch for renewables)
Government investment and consumption	Public investment	Public investment in climate R&D, energy or public transport infrastructure
	Public procurement	Public procurement of electric vehicles
	Public appraisal rules	Project and programme evaluation rules (e.g. requirement to conduct a Strategic Environmental Assessment before adopting a law)
Information instruments	Labels	Energy performance label for appliances, vehicles, etc.
	Reporting requirements	Energy audit requirement for industrial facilities
	Capacity building & public awareness	Public information campaign/education/training

[2] Instrument definitions are under development. “Voluntary GHG trading systems” refers to instruments that allow companies to opt into trading GHG allowances, implemented by governments e.g. to test the feasibility or effectiveness of a formal, mandatory ETS. Voluntary carbon markets, which are driven by the private sector, are beyond the scope of the IFCMA Climate Policy Database.

Voluntary approaches	Voluntary information instruments	Voluntary energy performance label Voluntary green investment label
	Voluntary trading systems	Voluntary GHG emissions trading system (e.g. pilot ETS)
	Voluntary targets	Voluntary GHG emission reduction target Voluntary energy efficiency target/ energy performance target

Source: Authors.

3 Data structure

A **data structure** is a coherent system that codifies information into organised data points. In the context of a policy database, it defines the unit of analysis, identifies the attributes necessary to either i) classify or ii) describe and characterise the various policy instruments countries have adopted. It also provides precise definitions of attributes and response options and, when feasible, standardises these response options. A well-designed data structure facilitates easy reporting or collection and updating, processing, retrieval and analysis of policy information, enabling policymakers, analysts, and researchers to gain insights, and support meaningful analysis and informed decision-making. As a general principle, the data structure should be flexible enough to accommodate evolving policy requirements and new data sources.

Overview of the data structure

The database is structured around the **policy instrument**, which is the primary unit of analysis (see Section 2 for the definition of a policy instrument). The database identifies policy instruments at a granular level, distinguishing them from broader policy packages. For example, a building code is commonly considered a policy instrument, but it typically includes multiple individual instruments that regulate different agents, assets or activities through various mechanisms (e.g. performance standards for windows or roofs; technology standards for heating systems, etc.). To ensure consistency of data collection and analysis, this database therefore focusses on distinct policy instruments as the unit of analysis, while also considering their aggregation into policy packages where they exist.

Information for each policy instrument is collected at two levels of aggregation: at the **policy instrument level** and at the **sub-scheme level,** enabling a more detailed description of the instrument and facilitating the use of the database for various needs, including GHG mapping.[3] Sub-schemes can be considered as variations of the main policy instrument. Often, these variations introduce differentiated requirements or provisions for specific segments of the policy regulatory base (to simplify *policy base*) (i.e. the group of agents, assets, or activities that an instrument's requirement applies to[4]) to address particular circumstances, industrial sectors, technologies or agents. For instance, a CO_2 emissions standard for motor vehicles may set distinct performance requirements for each vehicle class. Sub-schemes could also exempt certain assets or activities entirely (e.g. ambulances or miliary vehicles) or provide compensation or refunds. Sub-schemes can also help capture transitional periods (i.e. phase-in or phase-out mechanisms) and differentiated compliance requirements and enforcement mechanisms.

Each instrument and its sub-scheme is described through the policy **attributes** (in a matrix structure these are the columns, see Table 2). These attributes provide information on the nature, design, and status of the instrument.

[3] GHG emissions mapping is the practice of linking a policy instrument to the GHG emissions it directly targets or applies to (i.e. its emission base). See OECD (2024[3]) for a discussion.

[4] See OECD (2024[3]) and Bloc 3 in this section for a discussion of the policy base.

Table 2 Example of data structure

Policy instrument data structure					
	Attribute 1	*Attribute …*	*Attribute i*	*Attribute …*	*Attribute n*
Policy Instrument 1	*(1,1)*				
Sub-scheme 1: reduced rate					
Sub-scheme …					
Sub-scheme m: exemption					
Policy Instrument …					
Policy Instrument k			(k, i)		
Policy Instrument …					
Policy Instrument m					(m,n)

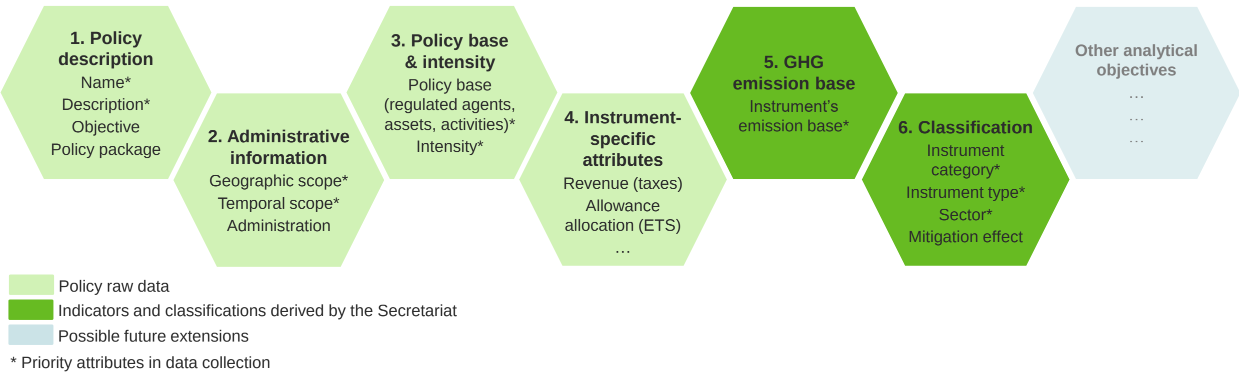
Source: Authors.

The selected attributes are defined with the aim of ensuring consistency with other OECD databases and environmental and climate policy projects, notably PINE, whenever feasible. This move towards harmonisation aims to prevent duplication by allowing for a unified data collection and validation process across various OECD programmes going forward. Additionally, attribute selection and definitions fully align with policy reporting guidelines of the ETF of the Paris Agreement (outlined in Annex F), should countries wish to draw on IFCMA policy data to support reporting to the UNFCCC.

To ensure a systematic structure and facilitate navigation, the policy attributes are grouped into six thematic blocs (Figure 3). The first four thematic blocs are considered “raw policy data”, or attributes that can be sourced from the legal statue(s) or regulation(s) establishing or detailing the instrument. Blocs five and six are information derived from the raw policy data by the IFCMA Secretariat, based on specific methodologies. The following section provides an overview of each bloc and the attributes captured therein. Annex C provides the full list of attributes, including their definitions and response options. As analytical or policy requirements evolve, new blocs could be integrated into the structure.

The attributes identified in this paper are comprehensive: they capture multiple dimensions of the design and implementation of policy instruments. This approach is intended to support various users and analytical purposes, opening new avenues for comparative and empirical policy analysis. However, given resource constraints, the first version of the IFCMA Climate Policy Database might not be able to include all attributes. Instead, certain attributes are identified as “priority attributes”, essential for the IFCMA’s GHG mapping and modelling; other attributes may be added later as resources permit, to measure climate policy design more broadly. Priority attributes are identified in Annex C.

Figure 3.1 Framework to structure policy instrument attributes



Source: Authors.

Bloc 1: Policy description

The policy description bloc provides general information about the policy instrument, including its **name** and a concise **description** of its main operating mechanism and specific design features (e.g. what the instruments imposes on, or provides for, whom and how). It also identifies the instrument's stated **objective**, when available, which also allows classifying an instrument as either a climate change mitigation or mitigation-*relevant* instrument, in line with IFCMA(2023)3/REV1, and to which legislative **policy package** the instrument belongs[5] (see Annex C for definitions of the attributes).

Bloc 2: Administrative information

The administrative information bloc captures the policy timeline, its geographical scope and information on administrative responsibilities. Concretely, it presents information on the instrument's **administrative level** (e.g. national vs. sub-national) and the **jurisdiction**. It also includes details related to the instrument's temporal scope, i.e. details on instrument **status** as well as **start date** and **end date**.[6] Note that, where a policy is implemented in phases (e.g. requirements becomes more stringent over time), each phase would be entered as a separate sub-scheme.

This bloc also includes the **administrating authority** that implements and manages the instrument. Collecting this information can support identification of experts potentially relevant to validate data on specific policy instruments. Additionally, it also identifies the legal document(s), and chapter(s) and/or article(s), that established or modified the instrument (if revised) as well as the weblink(s) to relevant government websites.

Bloc 3: Policy base and intensity

The third thematic bloc includes information related to the instrument's regulatory framework, namely the policy regulatory base (referred to as "policy base") and the specifics of the legal requirements imposed on regulated agents. The **policy base** refers to the combination of agents, activities, and assets subject to regulation under a given policy instrument, as defined by law ("de jure").[7] Capturing all exemptions or differentiated treatments of specific sub-categories of agents, activities or assets is pivotal to accurately defining the policy base, and a prerequisite for accurate mapping of policy instruments to their GHG emission base. Within the IFCMA data structure, such exemptions and differentiations are captured in sub-schemes.

The **regulated asset** is the item targeted by the policy instrument; this can be a structure (e.g. boilers and turbines), a fuel (e.g. coal or gas), emissions, more generic objects such as products and inputs, as well as intangible assets (e.g. income). In many cases, the asset determines the classification of instruments

[5] The "policy package" attribute is designed to capture 'vertical' linkages to broader regulatory frameworks, such as the United States' Inflation Reduction Act (IRA) of 2022 or the European Union's Fit-for-55 package, which contain several individual policy instruments. In addition, instrument-specific attributes (Bloc 4) capture selected 'horizontal' linkages, such as revenue reuse mechanisms or fees that apply when an entity is not in compliance with a regulatory instrument. Details on attributes provided in Annex C.

[6] IFCMA(2023)3/REV1 proposes to include policy instruments that are currently active or scheduled, meaning their implementation is planned to occur at a predefined future time (i.e. those that are enacted, but are not yet in force). Consequently, the database would encompass policies such as fuel economy standards, which, although enacted, do not require vehicle manufacturers' compliance with specific performance targets prior to a stipulated date, for instance, 1 January 2026. IFCMA(2023)3/REV1 furthermore suggests that the stocktake focus on national policy instruments, although some sub-national instruments will be part of the scope, considering their significance in certain jurisdictions.

[7] Also see OECD (2024[3])for a detailed discussion of the policy base.

to GHG emission source sectors (see Bloc 6). To increase consistency across entries in the database, facilitate filtering of instruments by regulated asset, and enable analysis across instruments and countries, regulated assets are standardised based on a pre-defined lists where possible (see Annex D for a non-exhaustive list of regulated assets relevant for the IFCMA database).

The **regulated agent** is the legal entity (person or institution) that must legally comply with the instrument and is liable for any non-compliance or receives the established benefits or information.[8] Regulated agents can include households, firms, governments, or non-governmental organisations, and further be characterised along several dimensions (e.g. households above/below a certain income threshold, firms above/below a certain production capacity, etc.). By capturing the regulated agent, it is possible to systematically capture, for example, whether a building regulation applies to public, commercial or residential buildings, or whether a subsidy benefits households or businesses.

The **regulated activity** refers to the actions or activities regulated by the policy instrument. These activities are usually defined by the legal obligation established by the instrument and carried out by the regulated agent. Activities may be identified to narrow down the policy base. For example, annual GHG emissions thresholds may be established as the minimum emissions necessary to qualify for a tax. Also a policy instrument may be defined in terms of activities or actions without specifying an asset or agent, for example restrictions, fines, or taxes on parking spaces or entering the city centre at certain times (see Bloc 6).

Another attribute is the instrument's **intensity**, which is specific to each instrument type. This attribute reports the legal obligation or benefit established for the regulated agent by the policy instrument. The requirement can be either qualitative or quantitative, which is more common. For quantitative obligations or benefits, often present for economic and regulatory instruments, the intensity attribute specifies the *level* of obligation or the benefit. For example, in a case of the carbon tax, it refers to the tax rate per tonne of CO_2e emitted; for an emissions standard, it provides the emission limit value; and for a feed-in-tariff, the payment received per unit of renewable electricity generated. Examples of instruments imposing qualitative requirements include information instruments requiring energy audits or framework standards requiring e.g. the adoption of Best Available Techniques (BAT).

As a general rule, the database will capture the intensity as stated in the regulatory statute establishing the instrument. For some instruments (e.g. taxes or emission standards), the intensity may be a single value, expressed potentially in different measurement units across jurisdictions.[9] However, in other cases, intensity values may be based on several dimensions, or formulas, which are more complex to record (an examples is a vehicle tax that is based on a formula considering value, weight, power train, pollution and efficiency performance). For instruments with a qualitative intensity (e.g. energy efficiency labels), intensity can be viewed as binary variables, i.e. required or not required. The Secretariat is exploring where and how standardisation can usefully enhance the transparent and consistent description of policies (see OECD (2024[6]) for an initial discussion)..

Bloc 4: Attributes specific to instrument types

While the attributes outlined above are pertinent for characterising any policy instrument, this bloc entails additional information relevant for specific instrument types to provide more precise information on the instrument's design or implementation. For instance, when considering a standard, a comprehensive description and assessment of its potential effectiveness may require information on enforcement

[8] Note that the agent that must legally comply with the instrument may not be the one that ultimately bears the financial burden of its implementation.

[9] For example, a fuel economy standard could be expressed in miles per gallon (mpg), litres per 100 kilometres (l/100 km), kilometres per litre (km/l), or grams of CO2 per kilometre (gCO2/km). In such cases, a simple unit conversion could standardise observations for enhanced transparency and cross-country analysis.

mechanisms, in addition to its intensity level,[10] and whether it allows for flexibility through mechanisms like averaging, banking and trading across affected entities. Conversely, for subsidies and taxes, attributes such as the distribution mechanism (i.e. by which mechanism a subsidy is awarded), or whether tax is earmarked may be of significance. These attributes may also capture direct links to other policies (e.g. revenue recycling to finance a specific low-carbon subsidy). Details on the instrument-specific attributes are provided in Annex C.

Bloc 5: GHG emission base

Bloc 5 and 6 are the two blocs that do not characterise an instrument through the attributes defined by law (de jure), rather it's the result of a methodology to relate climate policy data to emissions (de facto). Bloc 5 aims to quantify an instrument's GHG emission base, which determines which GHG emissions the instrument covers through its policy structure. The overarching concepts of how an instrument can be mapped to its emission base and the rationale for mapping are outlined in the IFCMA paper *Greenhouse Gas Emissions Mapping Methodology for Climate Change Mitigation and Mitigation-Relevant Policy* (OECD, 2024[3]).

The outcomes of such mapping can be expressed in different ways. Considering a carbon tax applied to emissions from stationary sources, the GHG emission base of the tax encompasses all CO_2 emissions originating from agents and operations liable under the tax. The emission base can either be denoted in terms of emissions covered in a particular year as an absolute figure (e.g. 5Mt of CO_2), or as a proportion of the total emissions of a sector (e.g. 70% of CO_2 emissions from power generation) or economy-wide emissions. The choice of the exact representation for the GHG mapping outcome will be shaped by the findings and experiences from the IFCMA pilot studies.

Bloc 6: Classification

Bloc 6 classifies each instrument along the policy instrument typology outlined in Section 2 (e.g. economic, regulatory, etc.). This classification is guided by the instrument's operating mechanism (e.g. does the instrument provide a direct financial benefit, does it price an activity, or does it impose a limit). This approach enables the systematic classification of instruments into types, based on the proposed IFCMA typology or relevant taxonomies as needed.

Furthermore, each instrument will be classified to sectors, including GHG **emissions source sectors** (based on the 2006 IPCC Guidelines for National Greenhouse Gas Inventories) and the **economic sectors** as outlined in the International Standard Industrial Classification of All Economic Activities (ISIC). The classification of the former typically depends on the assets regulated, while the latter depends on the regulated agent. The IPCC classification is important because many climate targets and emission trajectory analyses are based on it. The ISIC classification is relevant as many economic analyses and models rely on industry classifications like ISIC to represent economic activities and their interactions.

Finally, instruments will be tagged to relevant structural or functional characteristics, including their expected effect on GHG emissions,[11] as well as the GHGs and energy carriers likely to be affected. These classifications will be made by the Secretariat.

[10] Consider a GHG emission standard with an ineffective enforcement mechanism (such as low financial penalties in the case of non-compliance). Such a standard may not be effective in reducing GHG emissions as regulated agents are likely to simply accept to pay the penalty instead of performing the regulated activity.

[11] The expected effect on GHG emissions will categorise policies based on their likely impact on emissions – whether they are expected to increase emissions (e.g. fossil fuel subsidies) or reduce them (e.g. carbon tax). This classification, made by the Secretariat, reflects only the anticipated direction of the effect (positive or negative) and is not intended to provide a quantitative estimate of the actual change in physical emissions.

4 Summing up

This paper presents a comprehensive overview of the policy instrument typology and data structure for the IFCMA stocktake and the associated IFCMA Climate Policy Database. The typology is comprehensive and flexible, providing a harmonised framework for the classification and stocktake of climate policy instruments.

The data structure aims to organise policy information into consistent data points through key policy attributes, which are organised into coherent groups referred to as “blocs”. This structure not only allows for the coherent organisation of the information but also facilitates easy retrieval for analytical objectives.

The structure is comprehensive yet modular, allowing for the gradual and systematic coverage of policy instruments with their respective attributes in the future. Moreover, should analytical demands increase, additional attributes may be added, through blocs of information.

Finally, this structure allows for the coding of information, facilitating the development of information technologies for the storage, processing and administration of the information. The details of the definitions and classifications of both the policy instruments and their attributes are detailed in the Annexes.

References

Bemelmans-Videc, M., R. Rist and E. Vedung (1998), *CARROTS, STICKS and SERMONS: Policy Instruments and Their Evaluation*, https://doi.org/10.4324/9781315081748. [4]

Botta, E. and T. Koźluk (2014), "Measuring Environmental Policy Stringency in OECD Countries: A Composite Index Approach", *OECD Economics Department Working Papers*, No. 1177, OECD Publishing, Paris, https://doi.org/10.1787/5jxrjnc45gvg-en. [21]

D'Arcangelo, F. et al. (2022), "A framework to decarbonise the economy", *OECD Economic Policy Papers*, No. 31, OECD Publishing, Paris, https://doi.org/10.1787/4e4d973d-en. [14]

de Serres, A., F. Murtin and G. Nicoletti (2010), "A Framework for Assessing Green Growth Policies", *OECD Economics Department Working Papers*, No. 774, OECD Publishing, Paris, https://doi.org/10.1787/5kmfj2xvcmkf-en. [11]

Dubash et al. (2022), *National and sub-national policies and institutions. In IPCC,*, https://doi.org/10.1017/9781009157926.015. [5]

Duval, R. (2008), "A Taxonomy of Instruments to Reduce Greenhouse Gas Emissions and their Interactions", *OECD Economics Department Working Papers*, No. 636, OECD Publishing, Paris, https://doi.org/10.1787/236846121450. [10]

European Commission et al. (2009), *System of National Accounts 2008*, https://unstats.un.org/unsd/nationalaccount/docs/sna2008.pdf. [15]

Görlach, B. (2013), *What constitutes an optimal climate policy mix? Defining the concept of optimality, including political and legal framework conditions. CECILIA2050*, http://www.cecilia2050.eu. [12]

Goulder, L. and I. Parry (2008), "Instrument choice in environmental policy", *Review of Environmental Economics and Policy*, Vol. 2/2, pp. 152-174, https://doi.org/10.1093/reep/ren005. [13]

Gunningham, N., P. Grabosky and D. Sinclair (1998), *Smart Regulation*, Oxford University PressOxford, https://doi.org/10.1093/oso/9780198268574.001.0001. [7]

Gupta, S. et al. (2007), *Policies, Instruments and Co-operative Arrangements. In Climate Change 2007: Mitigation. Contribution of Working Group III to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*. [9]

IPCC (2023), “Summary for Policymakers”, in P.R. Shukla, J. (ed.), *Climate Change 2022 - Mitigation of Climate Change*, Cambridge University Press, Cambridge, UK and New York, NY USA, https://doi.org/10.1017/9781009157926.001. [1]

OECD (2024), “Greenhouse gas emissions mapping methodology for climate change mitigation and mitigation-relevant policy instruments: A methodological note to support GHG mapping for the IFCMA country pilot studies”, *Inclusive Forum on Carbon Mitigation Approaches Papers*, No. 2, OECD Publishing, Paris, https://doi.org/10.1787/373b72d3-en. [3]

OECD (2024), “Stocktaking and greenhouse gas mapping of transport policy instruments”, *Inclusive Forum on Carbon Mitigation Approaches Papers*, No. 3, OECD Publishing, Paris, https://doi.org/10.1787/247a5d21-en. [6]

OECD (2024), *The Policy Instruments for the Environment (PINE) Database, Methodology and definitions*, https://oe.cd/pinedatabase. [19]

OECD (2023), *Methodological Guidelines for Environmentally Related Tax Revenue Accounts*, OECD Publishing, https://doi.org/10.1787/d752d120-en. [22]

OECD (2023), “Policy Instruments for the Environment (PINE) Database”, Vol. June 2023 version, http://oe.cd/pine. [18]

OECD (2023), *Revenue Statistics 2023: Tax Revenue Buoyancy in OECD Countries*, OECD Publishing, Paris, https://doi.org/10.1787/9d0453d5-en. [16]

OECD (2023), *The Climate Action Monitor 2023: Providing Information to Monitor Progress Towards Net-Zero*, OECD Publishing, Paris, https://doi.org/10.1787/60e338a2-en. [2]

OECD (2008), *OECD Glossary of Statistical Terms*, OECD Publishing, Paris, https://doi.org/10.1787/9789264055087-en. [20]

Sorrell, S. et al. (2003), *Interaction in EU Climate Policy, Final Report to DG Research under the Framework V Project Interaction in EU Climate Policy*. [8]

United Nations et al. (2014), *System of Environmental Economic Accounting 2012 - Central Framework*, United Nations, New Tork, https://seea.un.org/sites/seea.un.org/files/seea_cf_final_en.pdf. [17]

Annex A. Review of typologies for environmental and climate change mitigation-related policy instruments

Policy instrument typologies are used either explicitly or implicitly in every analysis of policy instrument choice and design. This annex discusses selected typologies along the following criteria:

- **Comprehensiveness**: Covering the full range of instruments used to mitigate climate change, i.e. ensuring that all possible instruments fit into the typology.
- **Functional classification**: Organising policy based on their operating mechanism to facilitate analytical use. A well-defined operating mechanism outlines the rules that govern its implementation and the behaviour expected from the regulator and regulated agents.
- **Standardised unit**: Ensuring a comparable unit of observation (i.e., the instrument) to facilitate systematic data collection, statistical analysis, and the development of policy indicators. This criterion reflects the need to disentangle policy packages from policy instruments.
- **Explicit definitions**: Clear definitions with precise terminology to ensure common understanding of concepts and classifications and improve consistency, accuracy, and comparability of collected information.
- **Mutually exclusive and collectively exhaustive**: Any instrument should fit into one single category (mutually exclusive) and all possible instruments are covered by the categories (collectively exclusive), with no gaps. This is a result of the criteria outlined above that minimises room for interpretation.

The typologies discussed below have been chosen based on a comprehensive review of typologies that have been frequently cited or used in academic literature and by international organisation in the past two decades, prioritising comprehensive typologies (that aim to capture all possible environmentally relevant policy approaches), and those developed by governmental and intergovernmental organisations such as the OECD, the IPCC, and the European Commission. This is because many typologies which originate in academia tend to be designed and discussed in terms of policy effectiveness. The comprehensive nature of IFCMA aims to capture all policy approaches to mitigate climate change without preconceived notions of environmental effectiveness or economic efficiency. Typologies are reviewed by the year of publication, and the accompanying typologies are presented in their complete original form.

Bemelmans-Videc, Rist and Vedung (1998): Carrots, Sticks and Sermons: Policy instruments and Their Evaluation

The book "Carrots, Sticks, and Sermons: Policy Instruments and Their Evaluation" by Bemelmans-Videc, Rist and Vedung (1998[4]) provides a framework for classifying policy instruments into three broad categories: regulatory (sticks), economic (carrots), and informational (sermons). The classification is based on the mechanism through which policy instruments change behaviour: altering the relative costs of goods and services (*carrots* as economic instruments), mandating or restricting activities (*sticks* as regulatory instruments), or by providing information (*sermons* as information instruments). This framework has formed the foundation for numerous typologies in the literature, including those used by the IPCC and the UNFCCC reporting framework.

The IFCMA typology aligns with this framework but modifies and extends it by i) defining instrument types within categories, ii) ensuring that the unit of analysis in the typology is harmonised; iii) redefining economic and regulatory instrument to acknowledge that some instruments mix a carrot and stick mechanism (such as tradeable emission standards); and iv) making explicit the role of policy instruments related to government investment and consumption, information instruments and voluntary approaches as individual categories. This extension ensures the necessary breadth to cover the diverse set of instruments governments apply to mitigation climate changes, and the specific definitions needed for tracking policy instruments in a comprehensive and systematic manner.

Table A.1. Policy typology By Bemelmans-Videc, Rist and Vedung, 1998

First level
Carrots (Economic instruments)
Sticks (Regulatory instruments)
Sermons (Other)

Source: (Bemelmans-Videc, Rist and Vedung, 1998[4]).

Gunningham, Grabosky and Sinclair (1998): Smart Regulation: Designing Environmental Policy

The book “Smart Regulation: Designing Environmental Policy” by Gunningham, Grabosky and Sinclair (1998[7]) presents a framework for understanding and designing environmental policies. The authors introduce the concept of "smart regulation," which emphasises the strategic use of a mix of regulatory instruments to achieve environmental goals efficiently. As part of this effort, they construct a detailed typology of environmentally relevant policy instruments, categorising a wide range of instruments according to their operating mechanism, with detailed sub-instruments listed under each category.

However, the categorisation of certain instruments in the typology may result in inconsistent units of analysis. For instance, the category of "financial instruments" includes multiple sub-types, which may hinder the ability to compare attributes uniformly across all instruments. Additionally, the book provides detailed explanations and definitions of some yet not all instruments.

Table A.2. Policy typology by Gunningham, Grabosky and Sinclair, 1998

First level	Second level
Command and control regulation	Technology-based standards; Performance-based standards; Process-based standards
Self-regulation	
Voluntarism	
Education and information instruments	Education and training; Corporate environmental reports; Community right to know and pollution inventories; Product Certification; Award schemes
Economic instruments	Property-rights; Market creation; Fiscal instruments and charge systems; Financial instruments; Liability instruments; Performance bonds; Deposit refund systems; Removing perverse incentives
Free market environmentalism	

Note: The semicolon indicates that instrument types are mutually exclusive in the second level of the typology. “Self regulation” involves an industry-level organisation (as opposed to the government or individual firms) setting rules and standards (codes of practice) relating to the conduct of firms in the industry. “Voluntarism” is based on individual firms undertaking to do the right thing unilaterally, without any basis in coercion.
Source: (Gunningham, Grabosky and Sinclair, 1998[7]).

Sorrell et al. (2003): Interaction in EU Climate Policy, funded by the European Commission

The report "Interaction in EU Climate Policy, Final Report to DG Research under the Framework V Project Interaction in EU Climate Policy" by Sorell et al. (2003[8]), examines a wide range of EU climate policies. The report focuses on the potential interactions of policy instruments and aims at identifying potential overlaps and key factors, and developing recommendations based on these findings. As part of this attempt, a typology of climate change mitigation policy instruments is proposed based on earlier literature (Gunningham, Grabosky and Sinclair, 1998[7]).

(Sorrell et al., 2003[8]) offers an overview of policy instruments, detailing each instrument category with descriptions and examples. The paper lists a broad array of instruments and organises these instruments according to their operating mechanisms. However, it does not provide detailed definitions of the individual instrument mechanisms. A consistent unit of observation is only partly provided, as for instance "Education, information and moral suasion" could encompass various distinct instruments. The focus of the paper is on the interactions between the instruments rather than on the individual instruments themselves.

Table A.3. Policy typology by Sorrell et al., 2003

First level	Second level
Economic instruments	Charge systems; Trading mechanisms; Financial instruments
Command and control	Performance standards; Technology standards; Framework standards
Education, information, moral suasion	
Voluntary approaches	Unilateral commitments; Public voluntary schemes; Negotiated agreements

Note: The semicolon indicates that instrument types are mutually exclusive in the second level of the typology.
Source: (Sorrell et al., 2003[8]).

Gupta et al. (2007): Policies, Instruments, and Co-operative Arrangements, IPPC's Fourth Assessment Report

The IPCC's Fourth Assessment report includes a chapter on "Policies, Instruments, and Co-operative Arrangements" (Gupta et al., 2007[9]). This chapter provides a typology of policy instruments as part of a broader effort to synthesise information from the relevant literature, focusing on climate change mitigation policies. The instruments presented are evaluated according to four principles: environmental effectiveness, cost-effectiveness, distributional considerations, and institutional feasibility. Although the instruments are not divided into overarching categories, such as market-based-instruments and non-market-based instruments, they are explained and defined in groups according to their operating mechanisms.

The chapter provides an overview of a wide range of policy instruments categorised by their operating mechanisms. The report presents definitions, descriptions and examples of some, but not all, specific policy instruments. Furthermore, the report groups some instruments into categories which might not offer mutually exclusive units of observation. For instance, the typology includes a category on "Non-Climate policies", which diverts from the function classification applied in a first instance, and is not applied consistently throughout. In other words, the typology mixes the operating mechanism with the environment objective or outcome (i.e. intent or direct/indirect effect on GHG emissions).

Table A.4. Policy typology by Gupta et al., 2007

First level	Second level

Regulatory measures and standards	Technology standards; Performance standards
Taxes and Charges	
Tradable Permits	
Subsidies and other incentives	
Voluntary agreements	
Research and Development (R&D)	
Information instruments	Public disclosure requirements; Awareness/education campaigns; Labelling programmes; Rating and certification systems
Non-Climate policies	e.g. land use policies

Note: The semicolon indicates that instrument types are mutually exclusive in the second level of the typology.
Source: (Gupta et al., 2007[9]).

Duval (2008): A Taxonomy of Instruments to Reduce Greenhouse Gas Emissions and their Interactions, OECD working paper

The OECD working paper "A Taxonomy of Instruments to Reduce Greenhouse Gas Emissions and their Interactions" by (Duval, 2008[10]) serves as a primary source for other papers discussed below (e.g., (de Serres, Murtin and Nicoletti, 2010[11]) and (Görlach, 2013[12])). This paper focuses specifically on policy instruments for climate change mitigation, proposing a typology as part of a broader review of climate change mitigation policies and their interactions. Although it does not differentiate between broader categories, such as market-based instruments and non-market-based instruments, it groups the instruments by their operating mechanisms. Each instrument is assessed according to three broad cost-effectiveness criteria: static efficiency, level of innovation, and ability to cope with uncertainty. (Duval, 2008[10]) stands out as one of the few papers reviewed where the development of the typology for policy instruments is the principal objective.

The paper covers a range of relevant policy instruments and presents a categorisation along their respective operating mechanisms. Specific policy instruments are well defined and described in detail, offering valuable insight into their operating mechanisms. Moreover, there is a detailed comparison of instrument categories. Yet, it covers some instrument groups only partially, complicating the assessment of the standard units of analysis. For instance, the category "Technology-support policies" could include multiple instruments which could overlap with some economic instruments or reflect policy packages instead of instruments. Finally, instrument types such as information instruments are mentioned but neither included explicitly in the typology nor discussed in detail.

Table A.5. Policy typology by Duval, 2008

First level	Second level
GHG emission taxes	
Tradable emission permits	
Command-and-control approaches	Technology standards; Performance standards
Technology-support policies	
Voluntary agreements	

Note: The semicolon indicates that instrument types are mutually exclusive in the second level of the typology.
Source: (Duval, 2008[10]).

Goulder and Parry (2008): Instrument Choice in Environmental Policy

The academic paper "Instrument Choice in Environmental Policy" (Goulder and Parry, 2008[13]), published for the Association of Environmental and Resource Economists and the European Association of

Environmental and Resource Economists, discusses various environmental policy instruments and their implications, as well as the decision-making process involved in selecting pollution control instruments for environmental policy. The authors outline a comprehensive toolkit of environmental instruments. Additionally, the instruments are evaluated based on several criteria: economic efficiency (aggregate net benefits), cost-effectiveness, distribution of benefits or costs across different demographic groups, and the capacity to handle uncertainties.

As part of the analysis, the paper lists a range of different policy instruments and provides definitions for each of them. However, the range of instruments is not sufficiently comprehensive for the IFCMA. While the paper discusses research and development (R&D) investment policies and technology deployment policies, these are treated separately from other instrument categories. Information instruments and voluntary approaches are not covered. The unit of observation instruments remains somewhat ambiguous, as for instance technology deployment policies could take form of various instrument types, complicating their categorisation and analysis. Despite these limitations, the instruments covered are well defined and discussed in detail.

Table A.6. Policy typology by Goulder and Parry, 2008

First level	Second level
Incentive-Based Instruments	Emissions taxes; Tradable Emissions Allowances (Cap-and-Trade)
Direct Regulatory Instruments	Performance Standards; Mandates for Specific Technologies
Technology Policies	R&D policies; Technology deployment policies

Note: The semicolon indicates that instrument types are mutually exclusive in the second level of the typology.
Source: (Goulder and Parry, 2008[13]).

De Serres, Murtin and Nicoletti (2010): A Framework for Assessing Green Growth Policies, OECD Working Paper

De Serres, Murtin and Nicoletti (2010[11]) is an OECD Economics Department paper titled “A Framework for Assessing Green Growth Policies”. The primary objective of this paper is to propose an analytical framework for evaluating policies with a specific focus on integrating environmental externalities. The paper examines a broad spectrum of environmental policies and introduces a detailed typology of policy instruments. The typology categorises policy instruments into market-based-instruments and non-market-based instruments depending on their operating mechanisms, providing explicit lists of instrument types within each category. It serves as a foundational source for the typology proposed by (Görlach, 2013[12]).

The paper evaluates policy instruments based on several criteria, including cost-effectiveness, incentives for adoption and compliance, capacity to manage uncertainty, effectiveness in fostering innovation, and the facilitation of international coordination. The coverage of instruments, as well as the unit of analysis, are mostly aligning with the criteria chosen for the IFCMA typology. Although policy design characteristics are incorporated into this assessment, the paper does not offer detailed and explicit definitions for each instrument type, providing only an overview of the categories.

Table A.4. Policy typology proposed by de Serres, Murtin and Nicoletti, 2010

First level	Second level	Third level
Market instruments	Tax instruments	Taxes and charges directly applied to the pollution source; Taxes and charges applied on input or output of a production process causing environmental degradation
	Negative tax (or subsidy) for environmentally friendly activities	

	Deposit refund schemes	
Non-market instruments	Command-and-control (CAC) regulations	Technology standards; Performance standards; Bans; Obligations to obtain special permits and control-certificates; Reporting
	Active (green) technology-support policies	Public investment in R&D; Public investment in infrastructure; Public procurement
	Voluntary approaches	Rating and labelling programmes; Voluntary agreements

Note: The semicolon indicates that instrument types are mutually exclusive in the third level of the typology.
Source: (de Serres, Murtin and Nicoletti, 2010[11]).

Görlach (2013): What Constitutes an Optimal Climate Policy Mix? Defining the concept of optimality, including political and legal framework conditions, funded by the European Comission

The European Union's CECILIA2025 project aims to shed light on how policy instruments interact, what factors determine their performance, and how the European climate policy instrument mix should evolve to guide the transformation to a low-carbon economy. To support that aim, the paper "What Constitutes an Optimal Climate Policy Mix?", (Görlach, 2013[12]) synthesises a range of policy characteristics and delivers a comprehensive climate policy typology, including an extensive list of policy instruments. In this respect, it serves as a valuable base for the creation of the typology in the present paper. The report was published under a research project involving ten different academic institutions and funded by the European Union's 7th Framework Programme for Research. The typology in the paper, based on a comprehensive literature review, offers a detailed list of instruments, differentiating their characteristics, and provides definitions and selected examples for each instrument. Instruments are categorised into market-based and non-market-based, and further separated according to their operating mechanisms.

Based on this literature review, this typology provides the strongest foundation for the typology for the IFCMA. However, certain aspects would benefit from clearer sources and more detailed explanations of the criteria used to classify specific instruments. This is partly due to broad definitions which make it difficult to understand delimitations among mutually exclusive types. For instance, the classification of 'financial measures (subsidies)', 'feed-in-tariffs' and 'green certificates' as non-market-based instruments is ambiguous, as these typically influence the quantity or price of goods and services – a characteristic defining market-based instruments in many typologies. Similarly, a more detailed definition for instruments like 'framework standards' would enhance clarity regarding their source and operating mechanisms.

Table A.5. Policy typology by Görlach, 2013

First level	Second level	Third level
Market-based instruments		Taxes; Emissions trading systems; Removal of perverse incentives; Liability instruments; Deposit refund schemes
Non-market-based instruments	Command and control regulations	Framework standards; Performance standards; Technology standards; Prohibition or mandating of certain products or practices; Building codes and standards; Land use planning, zoning; Standalone reporting requirements; All (meaning reporting is part of all regulation policies)
	Active technology support policies	Public and private RD&D funding; Public procurement; Green certificates; Renewable portfolio standard; Feed-in tariffs; Public investment in underpinning infrastructure for new technologies
	Financial measures (subsidies)	Policies to remove financial barriers to acquiring green technology
	Information and voluntary approaches	Education and training; Product certification and labelling; Environmental labelling programs; Award schemes; Public information campaign; Voluntary agreements; Unilateral commitments; Public voluntary schemes

Note: The semicolon indicates that instrument types are mutually exclusive in the third level of the typology.
Source: (Görlach, 2013[12]).

Dubash et al. (2022): National and Sub-national Policies and Institutions, IPCC's Sixth Assessment Report

(Dubash et al., 2022[5]) is a chapter titled "National and Sub-national Policies and Institutions" included in the IPCC's "Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change" report of 2022. The report is aligned with seminal work on (Bemelmans-Videc, Rist and Vedung, 1998[4]) by grouping policy instruments along three blocs: economic instruments, regulatory instruments and other. The categorisation of specific policy instruments is based on empirical experience and acknowledges not to be comprehensive. This is because the typology is not the main objective of the chapter. Rather, the typology in (Dubash et al., 2022[5]) serves to provide a framework for the evaluation and assessment of policy instruments and packages. The criteria used for such assessment include environmental effectiveness, economic efficiency, distributional effects, side-effects, institutional requirements, and transformative potential.

Table A.6. Policy typology in Dubash et al.

First level	Second level
Economic instruments	Carbon taxes; Emission trading systems; Offset credits; Subsidies for mitigation; Removal of fossil fuel subsidies
Regulatory instruments	Performance standards, including tradable credits; Technology standards
Other policy instruments	Transition support policies; Information programmes; Public procurement and investment; Voluntary agreements

Note: Individual policy instruments in the third level are separated by a semicolon.
Source: (Dubash et al., 2022[5]).

D'Arcangelo et al. (2022): A Framework to decarbonise the economy, 2022, OECD working paper

The OECD working paper "A Framework to Decarbonise the Economy" (D'Arcangelo et al., 2022[14]) proposes a framework to design comprehensive decarbonisation strategies while promoting growth and social inclusion and to evaluate national climate targets and current policy approaches. It compares various policy instruments across criteria such as short-term (static) and medium-long term (dynamic) minimisation of abatement costs, administrative costs, ability to deal with uncertainty, distributional impacts, political economy and public acceptability, and fiscal implications.

The paper categorises policy instruments into three groups: emission pricing and other incentive-based instruments; regulations, standards, subsidies and hybrid instruments; and information instruments and voluntary approaches. Instruments are primarily classified based on the presence of a positive price on emissions, resulting in regulations and standards being grouped alongside subsidies. This contrasts with the IFCMAs objective to categorise instruments by operational mechanisms, distinguishing between instruments that incentive change by changing prices (e.g. through taxes, tax incentives, or per unit-subsidies and instruments that mandate specific actions or outcome, such standards.

The paper provides detailed definitions and descriptions of instruments within each category, enhancing the understanding of the typology's structure and facilitating comparative analysis across different policy approaches. However, the definition of the typology groups and the consistency of the units of analysis is not absolute. For example, subsidies and feebates are classified in the same group, as are technology and input standards, although the latter instruments could be considered sub-types of the former ones. Nonetheless, this paper serves as a valuable resource for constructing the IFCMA typology.

Table A.7. Policy typology in D'Arcangelo et al., 2022

First level	Second level
Emission pricing instruments and other incentive-based instruments	Greenhouse gas taxes; Emission trading systems (ETS); Taxes on polluting goods or activities
Regulations, standards, subsidies and hybrid instruments	Non-tradable performance standards/certificates; Abatement subsidies; Feebates; Technology standards; Input requirements; Tradable performance standards/certificates
Information and voluntary approaches	Rating and labelling requirements; Public information and education; Training programmes; Product certification; Award schemes

Note: The semicolon indicates that instrument types are mutually exclusive in the second level of the typology.
Source: (D'Arcangelo et al., 2022[14]).

Annex B. Definitions for the IFCMA policy typology

Economic instruments	Economic instruments refer to instruments that provide a direct monetary incentive on the levels of production or consumption, influencing prices or levels of production or consumption at the margin.
Subsidies	Subsidies are unrequited payments from government to non-government units.
Includes	Current unrequited payments on the basis of the levels of production or the quantities or values of the goods or services produced, consumed, sold or imported (European Commission et al., 2009[15]). Lump-sum payments from government to non-government units that are not based on quantities or prices (e.g. energy vouchers, agricultural land payments), even though they do not change prices at the margin. Performance-based subsidies that involve transactions from government to non-government units, where the latter provides a specific climate-relevant service in return for the payment, such as carbon sequestration or biodiversity protection, as is the case with payments for ecosystem services. Concessional loans through which a government unit lends to non-government units at intentionally below-market interest rates (European Commission et al., 2009[15]) as well as loan guarantees and other financial risk-mitigation mechanisms, if provided for goods or services produced, consumed, sold or imported. Feebates, i.e. systems in which a fee is levied on products or activities that produce high levels of GHG emissions, while a rebate is provided for those that produce low levels of emissions.
Excludes	Spending by a government unit towards forming fixed capital such as transport and energy infrastructure, or nature-based infrastructure solutions, as well as intangible assets such as research and development (R&D).
Examples	Electric vehicle purchase subsidies; feed-in-tariffs or premiums for renewable electricity; fossil fuel price ceilings (e.g. with compensation payments); motor vehicle feebates; payments for eco-system services.
Taxes	Taxes are compulsory unrequited payments to a government unit (OECD, 2023[16]) (United Nations et al., 2014[17])
Includes	Taxes with their standard rates as well as tax incentives within that tax regime. Examples include reduced energy excise tax rates for agriculture. Fees and charges, defined as compulsory requited payments to the government that are levied more or-less in proportion to the services provided.
Excludes	Penalties: payments conditional on non-compliance. Payments for licenses and permits: i.e. requitted payments. Feebates, i.e. systems in which a fee is levied on products or activities that produce high levels of GHG emissions, while a rebate is provided for those that produce low levels of emissions. Tax incentives for R&D activities.
Examples	Carbon tax; electricity excise tax; fuel excise tax; tax incentives for clean technologies; tax incentives for fossil fuel use; incineration tax; motor vehicle feebates.
Trading systems	Trading systems cause changes in the relative costs or prices faced by economic agents by regulating the trade of permits or certificates, which may cause a direct financial benefit (e.g. by providing the options of selling them) or set a price on the behaviour of the regulated agents (e.g. by providing the options of purchasing them).
Includes	Cap-and-trade emissions trading systems (ETS): sets a limit on GHG emissions by specified sources, distributes tradable allowances approximately equal to the limit, and requires regulated emitters to submit allowances equal to their verified emissions (Dubash et al., 2022[5]). Baseline-and-credit ETS: GHG emitters that reduce their emissions beyond a baseline level can earn

	credits that can then be sold to other agents that need them to meet their regulatory obligations. There is no fixed limit on GHG emissions. Any other tradeable standard, including for example, tradeable GHG and air emissions standards, tradeable renewable electricity generation certificates, or tradeable fuel economy standards. Offset credits: voluntary GHG emission reductions for which tradable credits are issued by a government body and can be used for compliance with a legal obligation (imposed e.g. by an ETS).
Excludes	Voluntary emission trading systems (e.g. pilot systems in which participation is not mandatory), which are classified under V*oluntary approaches*; offset systems not mandated or regulated by the government.
Examples	ETS cap-and-trade; ETS baseline-and-credit; tradable renewable energy obligations.
Regulatory instruments	Regulatory instruments define allowable actions or set limits to drive the adoption of desired processes, technologies, products, or outcomes. They may take the form of direct mandates, prohibitions, or restrictions on certain goods or activities, establish maximum or minimum limits on various parameters (such as emissions), or require agents to obtain explicit permission or meet specific criteria before engaging in particular activities. Failure to comply typically incurs sanctions, including monetary penalties.
Excludes	Regulatory instruments that mandate the provision of information (which are classified under *Information instruments*), instruments that regulate government spending and decision-making processes (which are classified under *Government investment and consumption*). Voluntary targets and activities (which are classified under *Voluntary approaches*). Standards where compliance certificates can be traded, such as tradeable GHG emission and air pollution standards or tradeable certificates from renewable electricity quotas (which are classified under *Trading systems*).
Performance standards	Performance standards mandate a specific outcome, giving regulated agents the freedom to choose technologies and processes to achieve the specified performance level.
Includes	Standards defined as a function of an output per unit of input. For example, energy efficiency regulations which mandate a minimum performance (e.g. hours of a working product per energy input such as brightness per hour per energy input, kilometres driven per fuel used, etc). Bans and phase outs: the instrument's outcome mandate implies a complete prohibition of specific inputs or technologies altogether (e.g. a fuel economy standard with an intensity of 0gCO_2/km).
Examples	GHG emission standards; air emission standards; fuel economy standards; minimum energy performance standards for electric equipment or buildings; speed limits; etc.
Technology standards	Technology standards require or prohibit a specific input or technology, or design for production.
Includes	Bans and phase outs: the instrument's input or technology requirement implies a complete prohibition of specific inputs or technologies altogether.
Examples	Biofuel blending standards; electric vehicles sales mandates; bans/ phase outs of fossil fuel-based home heating or power generation plants; etc.
Framework standards	Framework standards prescribe requirements on production processes without mandating the use of specific inputs or technologies or prescribing a specific outcome. Often, these requirements are of qualitative nature and they may require case-by-case interpretation and application. They may also regulate the functioning of markets.
Examples	In industry: Regulations defining the environmental permitting process, e.g. mandate the implementation of an energy or environmental management system (e.g. ISO 50001), or adoption of best available technology (BAT). In the electricity market: regulations that establish rules for market structure and guidelines for market participation, grid operation, and transmission system planning (e.g. priority grid access, dispatch, or curtailment based on environmental conditions or cost-efficiency). In land use: protected areas or zoning requirements.
Government investment and consumption	Government investment and consumption instruments encompass the allocation of financial resources by government units towards capital assets like infrastructure or research and development (R&D), as well as the implementation of policies and rules that shape government expenditure choices and decision-making processes.

Public investment	Public investment instruments refer to spending by a government unit towards forming fixed capital such as transport and energy infrastructure, or nature-based infrastructure solutions, as well as intangible assets such as R&D.
Includes	Government support provided to non-government units for R&D activities in the form of tax incentives. Government spending towards public transport and low-carbon mobility. Concessional loans, loan guarantees and other financial risk mechanisms, if provided for capital assets asses like infrastructure or R&D.
Excludes	Operating and maintaining costs that are not extending the assets lifetime. Lump-sum payments from government to non-government units which are included under *Subsidies*.
Public procurement	Public procurement refer to instruments that oblige or encourage public agents to consider climate change mitigation impacts when purchasing the goods, services and works from external suppliers or providers to fulfil public sector needs, deliver public services, and execute government projects and programmes.
Excludes	Budgetary transfers from government units to domestic non-government units related to public infrastructure and R&D activities, which are included under *Public Investment*; Education and trainings organised by government units.
Examples	Green public procurement rules.
Public appraisal rules	Public appraisal rules prescribe requirements on the decision-making processes and operations of the government without mandating the use of specific inputs or technologies or prescribing a specific outcome. Rather, they specify the process to be followed in decision-making processes and operations or define qualitative outcomes that may require case-by-case interpretation.
Examples	Requirements to conduct strategic environmental assessments on draft laws. Requirement to adopt an shadow carbon pricing in public decision-making.
Information instruments	Information instruments mandate the collection and/or dissemination of information to raise awareness and communicate knowledge to influence behaviour. These instruments typically promote voluntary technology choices and behavioural changes by firms and households, thereby allowing consumers to make better-informed choices.
Excludes	Instruments not subject to government oversight, such as private information campaigns (e.g., marketing, private labelling initiatives).
Labels	Mandate regulated agents to receive a label or certificate in exchange for information provision about the environmental performance, characteristics, or attributes of products, services, or activities.
Excludes	Voluntary labels and certifications (i.e. where firms can choose to participate), which are classified under *Voluntary approaches.*
Reporting requirements	Reporting requirements mandate agents to submit regular reports, disclosures, or documentation to the regulating agents, oversight bodies, or stakeholders.
Examples	Environmental auditing: requirements imposed on agents to conduct assessments or evaluations of their environmental performance, e.g. their energy consumption, efficiency, and performance.
Capacity building and public awareness	Government provides information to non-government units to disseminate information, raise awareness, and enhance understanding on specific issues, initiatives, or policies.
Includes	Public information campaigns and award schemes: systematic dissemination of information by government authorities or public agencies to raise awareness, educate the public, shape perceptions, influence attitudes, and promote behaviour change on specific issues or policy priorities. Capacity building, education and trainings: initiatives aimed at providing knowledge, skills, and learning opportunities to individuals, communities or organisations to enhance their understanding of specific issues, topics, or practices through formal government programmes or activities.
Voluntary approaches	Agreements between a government authority and one or more private parties to achieve climate objectives or to improve performance beyond compliance to regulated obligations. Especially for information instruments, they may also include frameworks developed by governments to facilitate private action (e.g. energy efficiency labels that firms may decide to use).
Excludes	Private agreements where the government is not involved.

Voluntary information instruments	Voluntary information instruments are established by governments to facilitate private action towards climate objectives. Participation is voluntary. Firms can choose to adopt to inform stakeholders and consumers about their environmental performance. Examples: voluntary energy efficiency labels; voluntary financial labels, voluntary reporting standards; voluntary audits.
Voluntary trading systems	Voluntary trading systems are established by government to allow organisations or companies to trade credits or allowances representing emissions reductions or other environmental benefits. Participation is voluntary, and these systems often operate parallel to or in the absence of mandatory trading systems. Examples: voluntary GHG emission trading systems.
Voluntary targets	Specific self-imposed goals set by organisations or companies to achieve climate objectives or improve environmental performance beyond regulatory compliance in collaboration with the government. Examples: voluntary GHG emission reduction targets, voluntary energy performance targets.

Annex C. Attributes to describe and characterise policy instruments in the IFCMA policy database

The first version of the IFCMA Climate Policy Database will not entail information for all attributes identified here. However, the aim of the data structure is to be comprehensive and provide the framework for a more detailed data collection that could unfold in the future. Priority attributes for the IFCMA stocktake, marked with asterisk (*), are necessary to conduct the GHG mapping and modelling envisioned under the IFCMA's technical work.

Any variation of the main policy instrument (i.e. a sub-schemes) is captured by differences in its attributes. For example, in the case of a carbon tax with rates that vary by fuel type, each sub-scheme would be represented by a distinct "regulated asset" attribute, along with its corresponding intensity.

Thematic bloc	Attribute	Description	Variable type	Response options
1. Policy description attributes	Domestic instrument name*	Instrument name in original language; preferably the name the policy is commonly known for (i.e. its colloquial name); if there is ambiguity about the instrument name, the name used in the instrument's legal statute(s)	Free text	N/A
	English instrument name*	The domestic instrument name translated to English	Free text	N/A
	Policy package*	The name of the policy package, if applicable. A policy package refers to an overarching framework that deliberately bundles and implements multiple instruments together to achieve synergistic effects, maximise overall impact, or minimise potential trade-offs. Examples include broad legislative packages such as building codes, or frameworks such as the United States 2022 Inflation Reduction Act or the European Union's Fit for 55 legislation. It excludes "horizontal" linkages between policies, such as earmarking tax revenue for specific subsidies; these linkages are captured through the instrument-specific attributes (Bloc 4)	Free text	N/A

	Description*	Brief description of the policy mechanism and specific design features	Free text	N/A
	Objective	Policy objective as stated in the legislation	Categorial (multiple choice)	Air pollution; Waste reduction/ circular economy; Climate change mitigation; Climate change adaptation; Economic growth; Energy efficiency; Energy security; Social security/poverty reduction; Other
2. Administrative information attributes	Country*	Country where the instrument is applied (not applicable for supra-/international instruments)	Categorial (single choice)	ISO 3-letter country code
	Jurisdiction level*	Jurisdiction level where the instrument is applied	Categorial (single choice)	National; Sub-national; Supra-national
	Jurisdiction name*	Name of jurisdiction if it is applied at supra- / international or subnational level	Free text	N/A
	Adoption date	Date when the instrument is approved	Fixed structure	[mm/yyyy] or [yyyy]
	Start date*	Date of entry into force/ when the instrument became or will become legally binding	Fixed structure	[mm/yyyy] or [yyyy]
	End date	Date when the final requirement of the legal mandate takes effect, if relevant	Fixed structure	[mm/yyyy] or free text if needed
	Last revisions*	Date of the last update/revision to the policy instrument	Fixed structure, free for details	[mm/yyyy] or [yyyy] or free for previous revisions
	Status*	Defines if the instrument is active or inactive, and whether it is in force	Categorial (single choice)	in force; scheduled; ended
	Administrating authority	The institution(s) that implement and manage the instrument. This involves setting rates and/or defining the policy base	Free text	N/A
	Legal statute	Name of relevant legal document(s) and chapter(s) and/or article(s) that established or modified the instrument (if revised)	Free text	N/A
	Legal document*	Weblink to the relevant legal document that established or modified the instrument (if revised)	Free text	N/A
	Other Weblinks	Weblinks to other relevant websites	Free text	N/A
3. Policy regulatory base and intensity	Policy base*	The combination of agents, activities, and assets that are regulated by a policy instrument, as defined the instrument's regulatory framework. The policy base determines the exact scope of the policy instrument, including who precisely has to comply with the instrument's legal obligation	N/A	N/A
	Regulated asset	The regulated asset is the object that is regulated; this can be a structure (e.g. boilers and turbines), a fuel (e.g. coal or gas), emissions, more generic objects	Categorial (multiple choice), free for details	Non-exhaustive list (see Annex D)

			such as products and inputs, as well as intangible assets		
	Regulated agent		Entity that must legally comply with the policy instrument, or that receive the established benefit or information	Categorial (multiple choice), free for details	Households; Firms; Governments; Non-governmental organisations
	Regulated activity		The point of incidence when the policy instrument intervenes	Categorial (multiple choice), free for details	Non-exhaustive list (see Annex D)
	Intensity*		The legal obligation imposed on, or the benefit provided for, the regulated agents. For many instrument types, this variable will capture a levels (e.g. tax rate, subsidy level, regulated thresholds etc.)	Numerical, free where needed	N/A
	Intensity Details		Further details about the intensity, if necessary, e.g. if the intensity is calculated on a formula, the formula and an average intensity value can be provided	Free text	N/A
4. Instrument-specific attributes	**Subsidies**	Annual expenditure	The financial allocation or budget earmarked in the year preceding the data collecting	Numerical	Value in domestic currency
		Subsidy level or rate*	The value of the subsidy, e.g. 10% of purchase price, USD 1000 per unit, etc.	Numerical	Value in domestic currency
		Limit	The overall financial allocation, budget earmarked over the entire duration of the instrument, the amount of assets subsidised (capacity limit), the time horizon of the subsidyor other limits on totals for the subsidy	Numerical	in domestic currency, USD or otherwise applicable unit
		Distribution mechanism	The mechanism by which the subsidy is awarded	Categorial	from list, e.g. auction type, by default, lottery, etc.
		Enforcement mechanism	If a performance is expected, how is non-compliance punished	Categorial, free for details	administrative action (e.g. warning, permit withdrawal); legal action (e.g. prosecution); monetary fines
		Subsidy floor or ceiling	The minimum or maximum limits set for subsidies per year (for the year the instrument is recorded), indicating the lower and upper bounds within which subsidy amounts must fall.	Numerical	Value in domestic currency
	Taxes [12]	Earmarked revenue*	The policy financed by the revenue of the recorded tax (recorded even if not otherwise part of this database)	Free text or binary	ID of the financed policy instrument or "Earmarked" if financed policy is not included in the database
		Annual revenue	Tax revenue of the recorded tax	Numerical	in domestic currency, USD or otherwise applicable unit
		Annual revenue forgone	If tax incentive, estimated loss in tax revenue due to the recorded tax incentive (recorded for the year the instrument is recorded)	Numerical	in domestic currency, USD or otherwise applicable unit

[12] Note that future iterations of this attribute table will add attributes capturing aspects of tax incentives, aligned with the OECD Investment Tax Incentives Database. Clarifications for these attributes are forthcoming in a future note.

	Trading systems	Type*	The trading system in place, i.e., trading of permits or credits.	Categorial	Cap-and-trade GHG emissions trading systems; Baseline-and-credit GHG emissions trading systems; Tradeable GHG emission standards; Offset credits; Other tradeable standards
		Cap	If a cap-and-trade system, the upper limit or cap imposed on the total greenhouse gas emissions allowed within a specific period under a GHG cap-and-trade or emissions trading system	Free text	N/A
		Allowance mechanism*	The system or method used to distribute or allocate emission allowances within the trading system, specifying how agents receive their allowances	Categorial (single choice)	E.g., Free allowance; Auctioning; Combination of both
		Revenue use	If the allowance distribution mechanism yields revenue, this attribute would record what the revenue is used for.		
		Volume	Traded volume in collection year-1	Numerical	N/A
		Linkages	Linkages with other ETS	Free	Enter the instrument IDs of relevant instruments (N/A if no link)
		Market stabilisation mechanism	The existence of any mechanism or provision within the trading system designed to stabilize the market, prevent extreme price fluctuations, or address imbalances in supply and demand. E.g., secondary markets/ offsets and credits	Dichotomous, free for details	yes/no
		Offset use allowed	Rules of which offsets are allowed or reference to such rules; "No" if no offsets allowed	Free	No if none; otherwise description of the applicable rules
		Penalties for non-compliance per unit	The consequences or fines imposed on agents that fail to comply with the regulations and requirements of the emissions trading system	Categorial, free for details	administrative action (e.g. warning, permit withdrawal); legal action (e.g. prosecution); monetary fines
	Regulatory	Compliance calculation methodology	Details about the design of the legal requirement (e.g. whether unit or fleet-based) and/or the approach/rules used to assess compliance with the instrument requirement (e.g. test cycle to measure emissions)	Free	N/A
		Compliance monitoring	The methods, processes, or systems in place to monitor and ensure compliance with the regulatory aspects of the climate change mitigation policy	Categorial (single choice), free for details	No compliance monitoring; Self-monitoring and reporting; Inspections or audits conducted by government authorities or third parties; Other
		Compliance enforcement	The consequences imposed on agents that do not adhere to the regulatory requirements of the climate change mitigation policy	Categorial (multiple choice), free for details	No compliance enforcement; Monetary penalties; Prohibitions; Compliance order; Other
		Compliance promotion*	Initiatives or activities aimed at encouraging and facilitating compliance with the regulatory measures of the climate change mitigation policy	Categorial (multiple choice), free for details	No compliance promotion; Regulatory incentives (e.g. weights, banking, pooling, trading); Information and

					direct assistance; Financial support; Other incentives or support
		Public investment: Co-financing	Indicates whether the government is financing the project/technology completely or is co-financing it	Categorial (single choice)	Yes; No
		Public investment: Selection process	Details regarding how the project or investment is chosen as an investment target. If there's a structured selection process, it indicates the criteria for selection	Categorial (single choice), free for details	Yes, No
		Public procurement: Tender process	Indicates the type of tender/ purchase process	Categorial (single choice), free for details	Open tender; Direct purchase; Negotiated purchase; Accelerated public tender
		Public procurement: Award criteria tender process	Indicates if there is a premium given to bidders that comply with specific requirements in the tender process	Free	N/A
	Government investment and procurement	Public procurement: Compliance monitoring	The methods, processes, or systems in place to monitor and ensure compliance with the regulatory aspects of the climate change mitigation policy	Categorial (single choice), free for details	No compliance monitoring; Self-monitoring and reporting; Inspections or audits conducted by government authorities or third parties; Other
		Public procurement: Compliance enforcement	The consequences imposed on agents that do not adhere to the regulatory requirements of the climate change mitigation policy.	Categorial (multiple choice), free for details	No compliance enforcement; Monetary penalties; Prohibitions; Compliance order; Other
		Public procurement: Life-cycle costing	Evaluating the financial proposals while considering the total costs over the life cycle of a product, service or works including (i) purchase price and all associated costs (delivery, installation, insurance, etc.); (ii) operating costs, including energy, fuel and water use, spares, and maintenance (iii) end-of-life costs, such as decommissioning or disposal; and (iv) costs imputed to environmental externalities.	Categorial (single choice)	Yes; No
		Responsibility information capturing	Details on who is responsible for capturing the required information.	Categorial (single choice)	Government authorities (inspections or audits); Third parties (inspections or audits); The regulated agents (self-reporting)
		Information transmission	Indicates how the information is transmitted to the government.	Free	N/A
		Frequency of information provision	Indicates how often the information has to be provided.	Categorial (single choice), free for details	In real-time; Quarterly; Annually; Other, details:__
	Information	Public availability of information	Details about whether the information is publicly available and, if so, where it can be accessed.	Free	N/A
		For labels: Type	Indicates how labels are assigned and designed.	Categorial	Binary (based on a single threshold value); Categorial; Continuously
		Compliance monitoring	The methods, processes, or systems in place to monitor and ensure compliance with the regulatory aspects of the climate change mitigation policy.	Categorial (single choice), free for details	No compliance monitoring Random; Targeted, according to a criteria:__; Regular temporal intervals:__; Irregular temporal intervals;

					Inspections/audits with prior notice; Inspections/audits without prior notice
		Compliance enforcement	The consequences imposed on agents that do not adhere to the regulatory requirements of the climate change mitigation policy.	Categorial (multiple choice), free for details	No compliance enforcement; Monetary penalties; Prohibitions; Compliance order; Other
		Compliance promotion	Initiatives or activities aimed at encouraging and facilitating compliance with the regulatory measures of the climate change mitigation policy.	Categorial (multiple choice), free for details	No compliance promotion; Information and direct assistance; Financial support; Other incentives or support
		Details on capacity building, training/education, information campaigns and award schemes	Indicates details on the frequency, the target group and the annual budget allocated	Free	N/A
	Voluntary approaches	Incentives for Participation	The rewards, benefits, or positive inducements offered to encourage voluntary participation in climate change mitigation activities or programs.	Categorial (single choice)	Financial; Regulatory; Reputational; None
		Monitoring	The processes and systems in place to track and assess voluntary actions or initiatives related to climate change mitigation.	Dichotomous, free for details	Yes/no, free for detail
		Sanctions for non-compliance	The consequences or penalties imposed on agents or individuals who fail to fulfill their voluntary commitments or obligations related to climate change mitigation.	Dichotomous, free for details	Yes/no, free for detail
5. GHG emission base *(developed by Secretariat)*	GHG emission base*		The total GHG emissions the instrument targets or potentially affects. It is defined by the instrument's policy base and the emission source, which may or may not be within the policy base (OECD, 2024[3]). Could be expressed in absolute of relative terms (e.g. as % of a sector).	Numerical	N/A
6. Classification *(developed by Secretariat)*	Instrument Category*		Classification based on IFCMA typology presented in Section 2	Categorial (single choice)	Economic; Regulatory; Government investment and consumption; Information instruments; Voluntary approaches
	Instrument Type*		Classification based on IFCMA typology presented in Section 2	Categorial (single choice)	Subsidies; Taxes; Trading systems; Performance standards; Technology standards, etc.
	Instrument*		Classification into instruments based on non-exhaustive list	Categorial (single choice)	Non-exhaustive list (to be developed)
	Economic sector*		Classification based on the regulated agent. If the regulated agent is a firm, the sector will be allocated according to the International Standard Industrial Classification of All Economic Activities (ISIC)	Categorial (multiple choice)	Households; Firm; Government, Non-government; if firm, sub-classification into economic sector according to ISIC (see Annex E)
	Emissions sector*		The emissions source sector in which the instrument affects GHG emissions. Classification based on the	Categorial (multiple choice)	Agriculture, Forestry and other land use (AFOLU); Energy use (Agriculture; Buildings; Manufacturing and

	technology the instrument addresses, which is often determined through the regulated asset		construction; Transport; Energy industries; Other); Waste, Industrial processes and product use (IPPU)
Mitigation effect	Describes whether the expected mitigation effect is likely to be is positive, negative, neutral or unknown	Categorial (single choice)	positive, negative, neutral, unknown
Mitigation co-benefits	Additional, non-mitigation positive outcomes or advantages that result from the implementation of climate change mitigating measures, identified by the IFCMA Secretariat.	Categorial (multiple choice)	Adaptation; Air pollution; Energy supply security; Technological innovation; Poverty reduction; Employment; Industrial development; Other (non-exhaustive list, current options here based on IPCC)
GHGs affected*	GHGs that are regulated (where directly regulated); otherwise GHG that are likely to be affected by the instrument.	Categorial (multiple choice)	CO_2; CH_4; N_2O; SF_6; HFCs; PFCs; NF_3

Annex D. Preliminary list of regulated assets and activities

The list of regulated assets provides reference options to facilitate the harmonisation of information. It is based on the UNSD Central Product Classification (CPC) Version 2.1, but amended to better capture climate-relevant assets, i.e. directly or indirectly affects GHG emissions or is influenced by climate change-related factors. The assessment on what constitutes a climate-relevant asset was guided by the assets that are linked to emission sources and activities listed in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. The list below also includes low- or zero-emitting assets that can replace emitting assets as well as assets that may cause indirect emissions in the power sector by consuming electricity (e.g. electric appliances).

The list is non-exhaustive and deemed to evolve over time, e.g. during the pilot phase of the IFCMA.

List of regulated assets

Motor vehicles, trailers and semi-trailers and its parts and equipment
Road transport vehicles
Cars

Trailers and semi-trailers
Light-duty trucks
Heavy-duty trucks
Busses
Motorcycles
Aquatic hips
Railway and tramway locomotives and rolling stock
Aircraft
Other transport equipment
Constructions
Buildings
Power plants
Industrial plants
Residential buildings
Commercial buildings
Public buildings
Parts of buildings and constructions (windows, doors, and others)
Buildings energy systems (electricity systems, heating systems, etc)
Space heating systems
Space cooling systems
Water heating systems
Cooking systems
Lighting systems
Road infrastructure
Rail infrastructure
Port infrastructure
Electricity grid
Pipelines
Mines
Land
Forest Land
Cropland
Grassland

Wetlands
Settlements
Other land

Machinery and equipment

Engines and turbines and parts thereof
Pumps, compressors, hydraulic and pneumatic power engines, and valves, and parts thereof
Ovens and furnace burners and parts thereof
Lifting and handling equipment and parts thereof
Gas generators; distilling plant; filtering machinery
Air-conditioning and refrigerating equipment
Agricultural or forestry machinery and parts thereof
Machinery for mining, quarrying and construction, and parts thereof
Domestic appliances and parts thereof
Electrical machinery and apparatus
Accumulators, primary cells and primary batteries, and parts thereof
Other machinery and equipment
Ores and minerals; energy, electricity, gas and water products
Energy carriers
Coal
Coke oven products; refined petroleum products; nuclear fuel
Coke and semi-coke of coal, of lignite or of peat; retort carbon
Tar distilled from coal, from lignite or from peat, and other mineral tars
Peat
Natural gas
Electricity
Heat
Oil
Crude petroleum
Motor gasoline
Aviation gasoline
Naphtha
Kerosenes
White spirit and special boiling point industrial spirits
Gas oil
Fuel oils (not elsewhere classified)
Lubricants
Other petroleum oils (not elsewhere classified)
Propane and butanes, liquefied
Ethylene, propylene, butylene, butadiene and other petroleum gases or gaseous hydrocarbons, except natural gas
Biofuels
Biomass
Waste
Industrial waste

Municipal waste
Nuclear energy
Solar energy
Wind energy
Hydropower
Geothermal
Other renewable energies
Ozone depleting substances
Substitutes of Ozone depleting substances
Others
GHGs
R&D outputs (studies, papers, etc.)
Information

List of regulated activities

Conservation, protection or storage
Extraction, abstraction or harvesting
Production, generation or conversion
Abatement or prevention
Export, import, sales or purchase
Registration, licencing or other administrative tasks
Consumption (use, ownership or capital formation)
Disposal, collection or sorting after use
Recycling, repurposing or other treatment after use
Research and development

Annex E. Classification to economic activities

Based on the International Standard Industrial Classification of All Economic Activities (ISIC), Rev. 4

Regulated agents
Households
Government
Other non-government
Firms
A - ALL - Agriculture, forestry and fishing
A01 - Crop and animal production, hunting and related service activities
A02 - Forestry and logging
A03 - Fishing and aquaculture
B - ALL - Mining and quarrying
B05 - Mining of coal and lignite
B06 - Extraction of crude petroleum and natural gas
B07 - Mining of metal ores
B08 - Other mining and quarrying
B09 - Mining support service activities
C - ALL - Manufacturing
C10 - Manufacture of food products
C11 - Manufacture of beverages
C12 - Manufacture of tobacco products
C13 - Manufacture of textiles
C14 - Manufacture of wearing apparel
C15 - Manufacture of leather and related products
C16 - Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials
C17 - Manufacture of paper and paper products
C18 - Printing and reproduction of recorded media
C19 - Manufacture of coke and refined petroleum products
C20 - Manufacture of chemicals and chemical products
C21 - Manufacture of basic pharmaceutical products and pharmaceutical preparations
C22 - Manufacture of rubber and plastics products
C23 - Manufacture of other non-metallic mineral products
C24 - Manufacture of basic metals
C25 - Manufacture of fabricated metal products, except machinery and equipment
C26 - Manufacture of computer, electronic and optical products
C27 - Manufacture of electrical equipment
C28 - Manufacture of machinery and equipment n.e.c.
C29 - Manufacture of motor vehicles, trailers and semi-trailers
C30 - Manufacture of other transport equipment
C31 - Manufacture of furniture
C32 - Other manufacturing
C33 - Repair and installation of machinery and equipment
D - ALL - Electricity, gas, steam and air conditioning supply
D35 - Electricity, gas, steam and air conditioning supply
E - ALL - Water supply; sewerage, waste management and remediation activities
E36 - Water collection, treatment and supply
E37 - Sewerage
E38 - Waste collection, treatment and disposal activities; materials recovery
E39 - Remediation activities and other waste management services
F - ALL - Construction
F41 - Construction of buildings
F42 - Civil engineering
F43 - Specialized construction activities
G - ALL - Wholesale and retail trade; repair of motor vehicles and motorcycles
G45 - Wholesale and retail trade and repair of motor vehicles and motorcycles
G46 - Wholesale trade, except of motor vehicles and motorcycles
G47 - Retail trade, except of motor vehicles and motorcycles
H - ALL - Transportation and storage
H49 - Land transport and transport via pipelines
H50 - Water transport
H51 - Air transport
H52 - Warehousing and support activities for transportation
H53 - Postal and courier activities
I - ALL - Accommodation and food service activities
I55 - Accommodation
I56 - Food and beverage service activities
J - ALL - Information and communication
J58 - Publishing activities
J59 - Motion picture, video and television programme production, sound recording and music publishing activities
J60 - Programming and broadcasting activities
J61 - Telecommunications
J62 - Computer programming, consultancy and related activities
J63 - Information service activities
K - ALL - Financial and insurance activities
K64 - Financial service activities, except insurance and pension funding
K65 - Insurance, reinsurance and pension funding, except compulsory social security
K66 - Activities auxiliary to financial service and insurance activities
L - Real estate activities
M - ALL - Professional, scientific and technical activities
M69 - Legal and accounting activities
M70 - Activities of head offices; management consultancy activities
M71 - Architectural and engineering activities; technical testing and analysis
M72 - Scientific research and development
M73 - Advertising and market research
M74 - Other professional, scientific and technical activities
M75 - Veterinary activities
N - ALL - Administrative and support service activities
N77 - Rental and leasing activities
N78 - Employment activities
N79 - Travel agency, tour operator, reservation service and related activities
N80 - Security and investigation activities
N81 - Services to buildings and landscape activities
N82 - Office administrative, office support and other business support activities
O - Public administration and defence; compulsory social security
O85 - Public administration and defence; compulsory social security
P - Education
P85 - Education
Q - ALL - Human health and social work activities
Q86 - Human health activities
Q87 - Residential care activities
Q88 - Social work activities without accommodation
R - ALL - Arts, entertainment and recreation
R90 - Creative, arts and entertainment activities
R91 - Libraries, archives, museums and other cultural activities
R92 - Gambling and betting activities
R93 - Sports activities and amusement and recreation activities
S - ALL - Other service activities
S94 - Activities of membership organizations
S95 - Repair of computers and personal and household goods
S96 - Other personal service activities
T - ALL - Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use
T97 - Activities of households as employers of domestic personnel
T98 - Undifferentiated goods- and services-producing activities of households for own use
U - Activities of extraterritorial organizations and bodies
U99 - Activities of extraterritorial organizations and bodies

Annex F. Reporting framework under the UNFCCC

UNFCCC reporting processes for information on parties' policies and measures are guided by the Modalities, procedures and guidelines (MPGs) for the transparency framework for action and support referred to in Article 13 of the Paris Agreement (Decision 18/CMA.1) as well as the guidance on operationalizing the MPGs as per Decision 5/CMA.3.

The common tabular format for the reporting of mitigation policies and actions necessary to track progress made in implementing and achieving nationally determined contributions under Article 4 of the Paris Agreement is provided in Table 5 of Annex II of Decision 5/CMA.3.

5. Mitigation policies and measures, actions and plans, including those with mitigation co-benefits resulting from adaptation actions and economic diversification plans, related to implementing and achieving a nationally determined contribution under Article 4 of the Paris Agreement[a,b]

Name[c]	*Description[d,e,f]*	*Objectives*	*Type of instrument[g]*	*Status[h]*	*Sector(s) affected[i]*	*Gases affected*	*Start year of implementation*	*Implementing entity or entities*	*Estimates of GHG emission reductions (kt CO_2 eq)[j,k]*	
									Achieved	*Expected*

[a] Each Party shall provide information on actions, policies and measures that support the implementation and achievement of its NDC under Article 4 of the Paris Agreement, focusing on those that have the most significant impact on GHG emissions or removals and those impacting key categories in the national GHG inventory. This information shall be presented in narrative and tabular format (para. 80 of the MPGs).
[b] For each Party with an NDC under Article 4 of the Paris Agreement that consists of mitigation co-benefits resulting from Parties' adaptation actions and/or economic diversification plans consistent with Article 4, para. 7, information to be reported under paras. 80, 82 and 83 of the MPGs includes relevant information on policies and measures contributing to mitigation co-benefits resulting from adaptation actions or economic diversification plans (para. 84 of the MPGs).
[c] Parties may indicate whether a measure is included in the 'with measures' projections.
[d] Additional information may also be provided on the cost of the mitigation actions, non-GHG mitigation benefits, and how the mitigation action interacts with other mitigation actions, as appropriate (para. 83(a–c) of the MPGs)
[e] Parties should identify actions, policies and measures that influence GHG emissions from international transport (para. 88 of the MPGs).
[f] Parties should, to the extent possible, provide information about how actions, policies and measures are modifying longer-term trends in GHG emissions and removals (para. 89 of the MPGs).
[g] Parties shall, to the extent possible, provide information on the types of instrument: regulatory, economic instrument or other (para. 82(d) of the MPGs).
[h] Parties shall, to the extent possible, use the following descriptive terms to report on status of implementation: planned, adopted or implemented (para. 82(e) of the MPGs).
[i] Parties shall, to the extent possible, provide information on sector(s) affected: energy, transport, industrial processes and product use, agriculture, LULUCF, waste management or other (paras. 81 and 82(f) of the MPGs).
[j] Each Party shall provide, to the extent possible, estimates of expected and achieved GHG emission reductions for its actions, policies and measures in the tabular format; those developing country Parties that need flexibility in the light of their capacities with respect to this provision are instead encouraged to report this information (para. 85 of the MPGs).
[k] To the extent available, each Party shall describe the methodologies and assumptions used to estimate the GHG emission reductions or removals due to each action, policy and measure. This information may be presented in an annex to the biennial transparency report (para. 86 of the MPGs).

Source: UNFCCC (2021), Decision 5/CMA.3 Guidance for operationalizing the modalities, procedures and guidelines for the enhanced transparency framework referred to in Article 13 of the Paris Agreement, CMA2021_L10a2E.pdf (unfccc.int).

For more information:

www.oecd.org/climate-change/inclusive-forum-on-carbon-mitigation-approaches

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